

Authorized update to MUD URLs

Michael Richardson
IETF123 OPSAWG meeting
July 24, 2025
(last time was in opsawg)

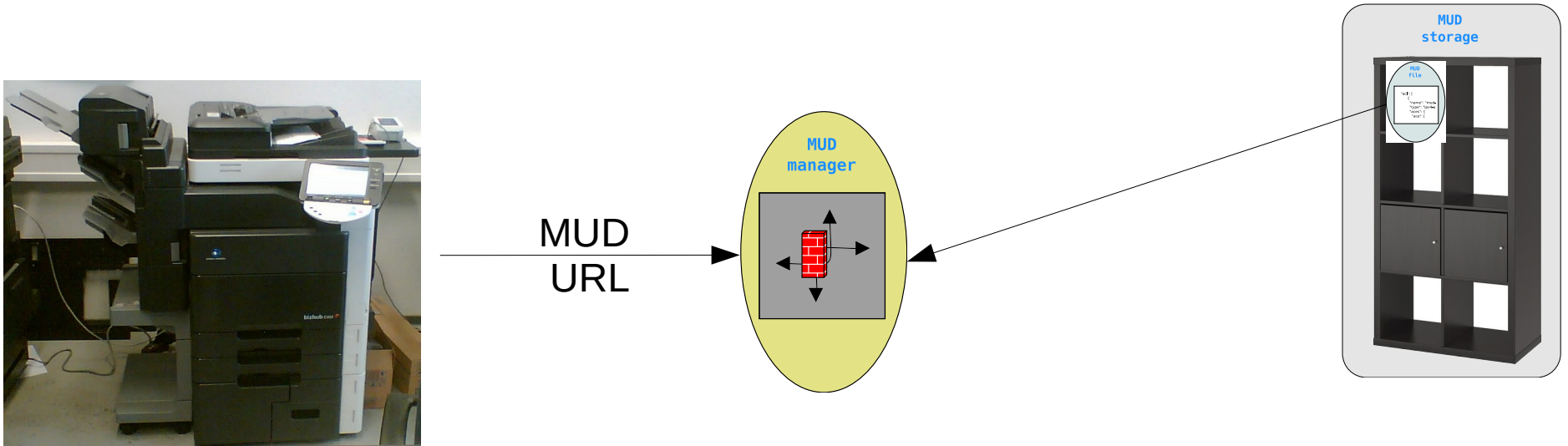
`draft-ietf-iotops-mud-acceptable-
urls-00`
(was `draft-ietf-opsawg-mud-
acceptable-urls-13`)



MUD URLs vs MUD files

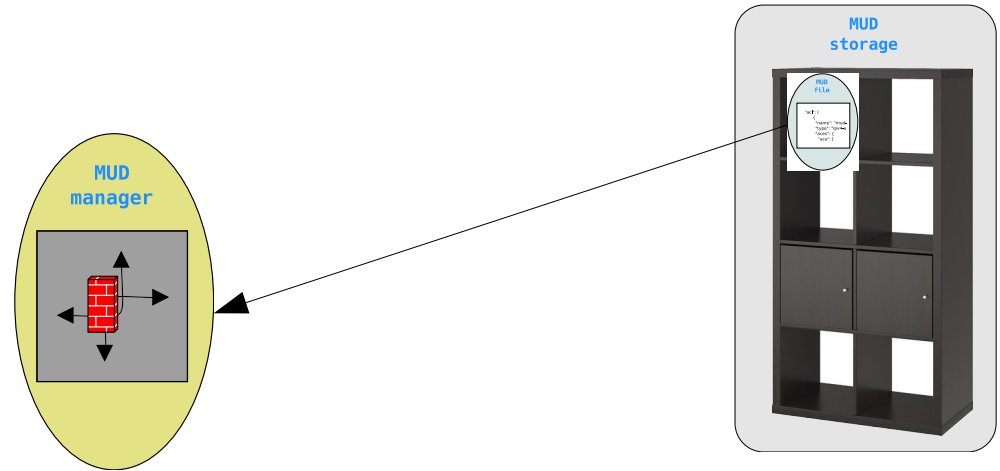
- Update MUD file in place
 - PRO: no security problems
 - CON: opens issues with mis-matches of firmware
 - e.g., TLS profiles!
- Update MUD URL, provide new MUD file
 - PRO: every major revision can have specific rules
 - PRO: different operating modes for device can express different preferences
 - (commercial, pro-user, residential)
 - Features enabled via accessories
 - CON: risk that MUD file link could be changed by malware

What is the problem?



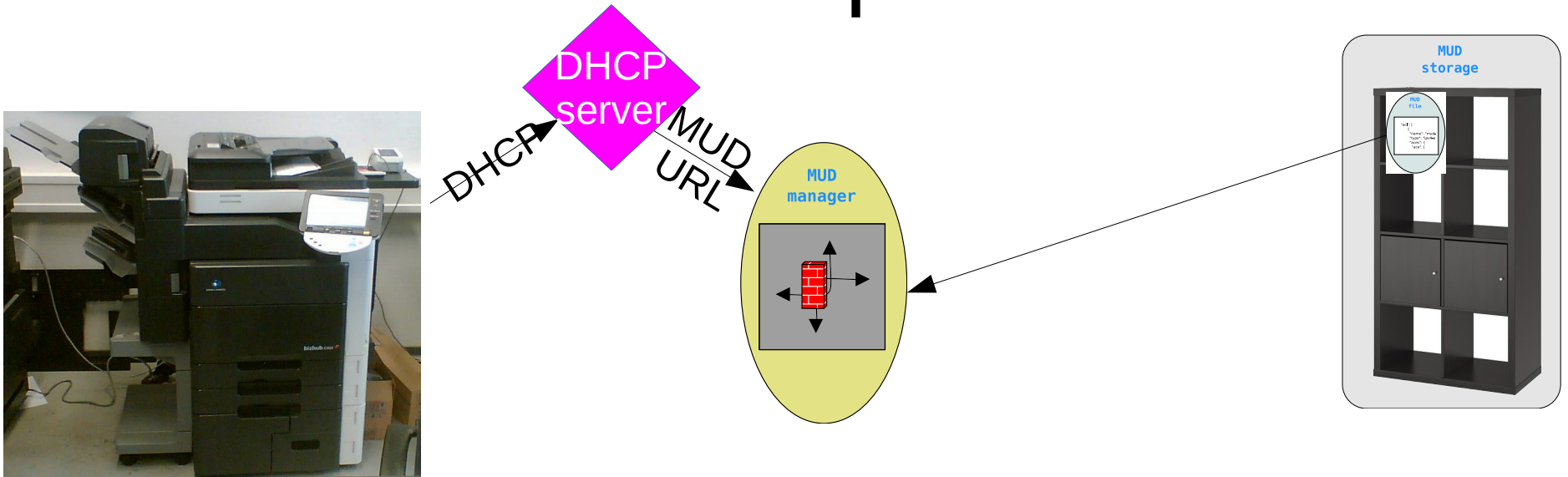
- IoT device tells MUD manager about it's URL
 - MUD manager fetches the file

What is the problem?



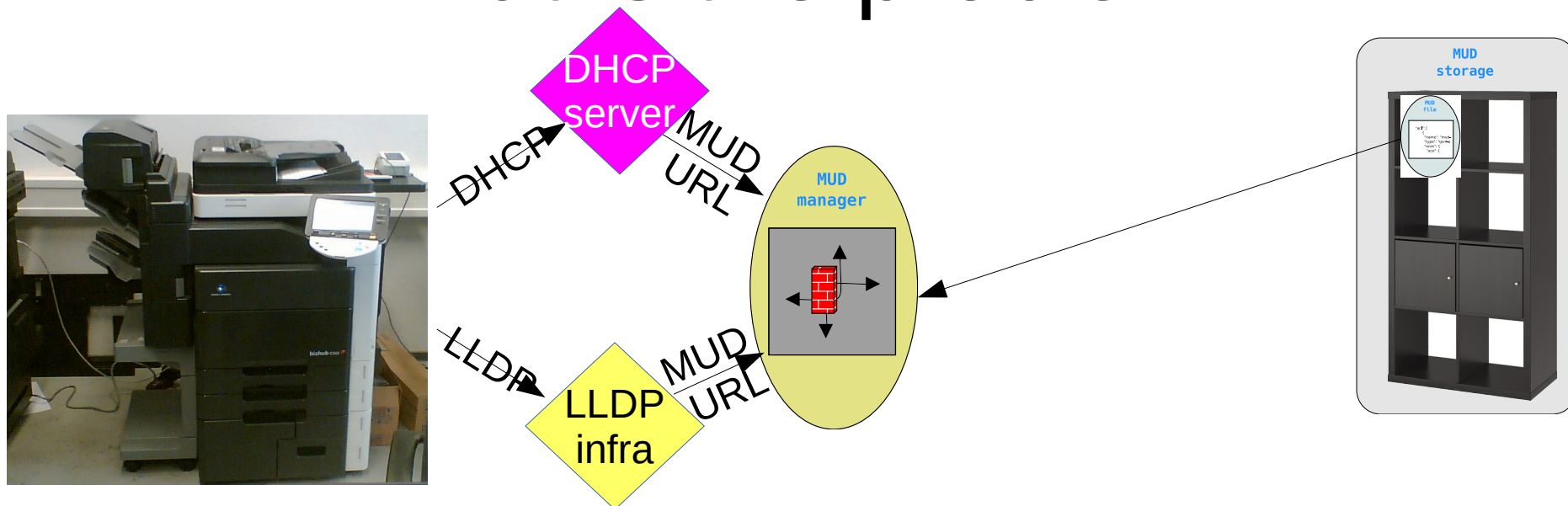
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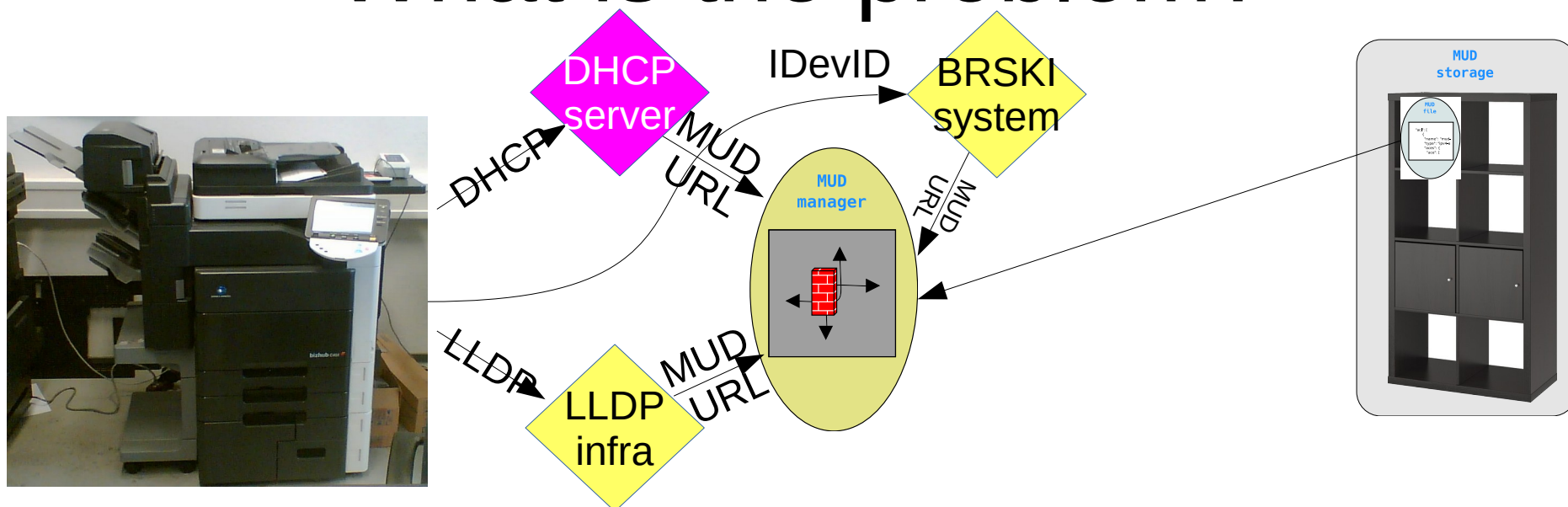
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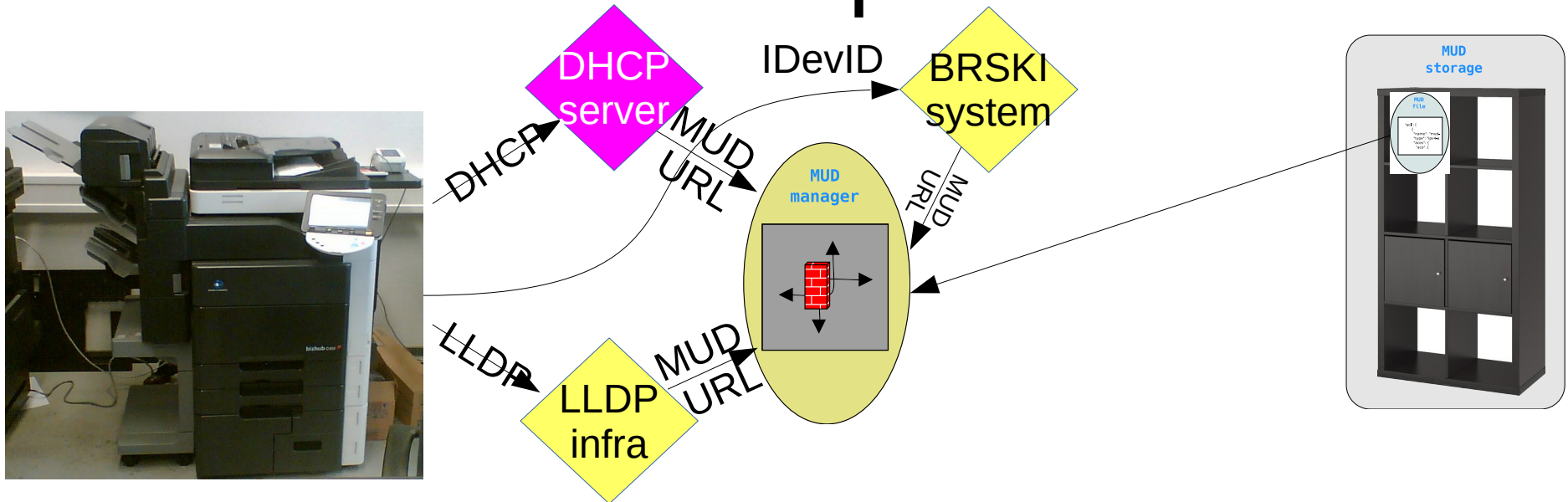
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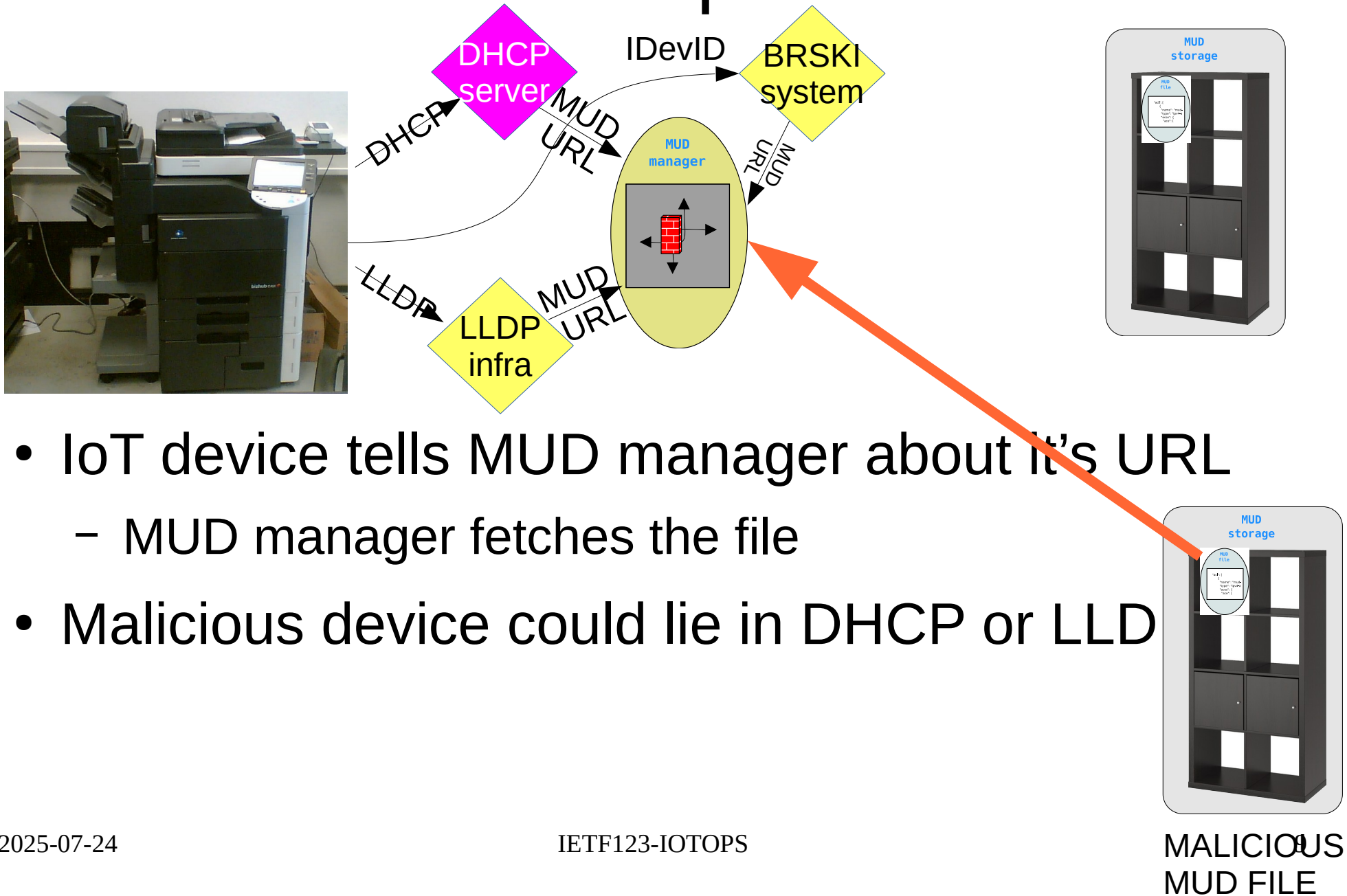
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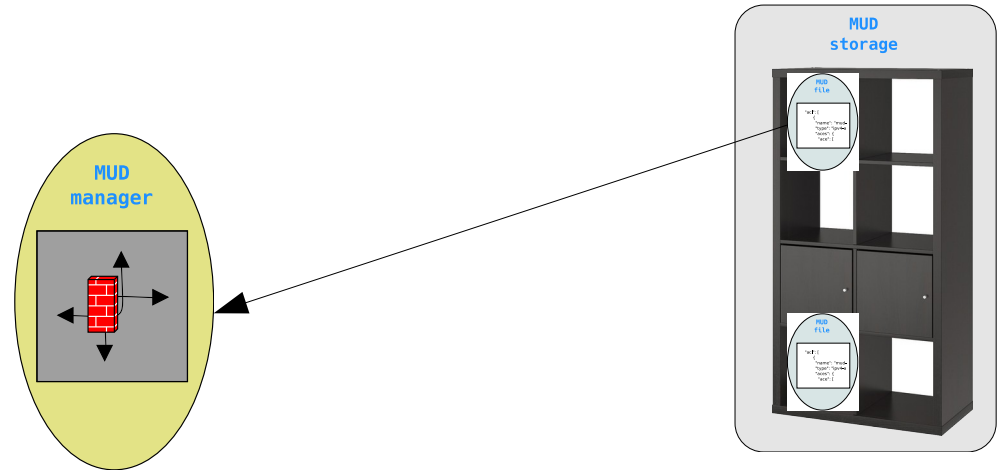


- IoT device tells MUD manager about it's URL
 - MUD manager fetches the file
- Malicious device could lie in DHCP or LLDP

What is the problem?

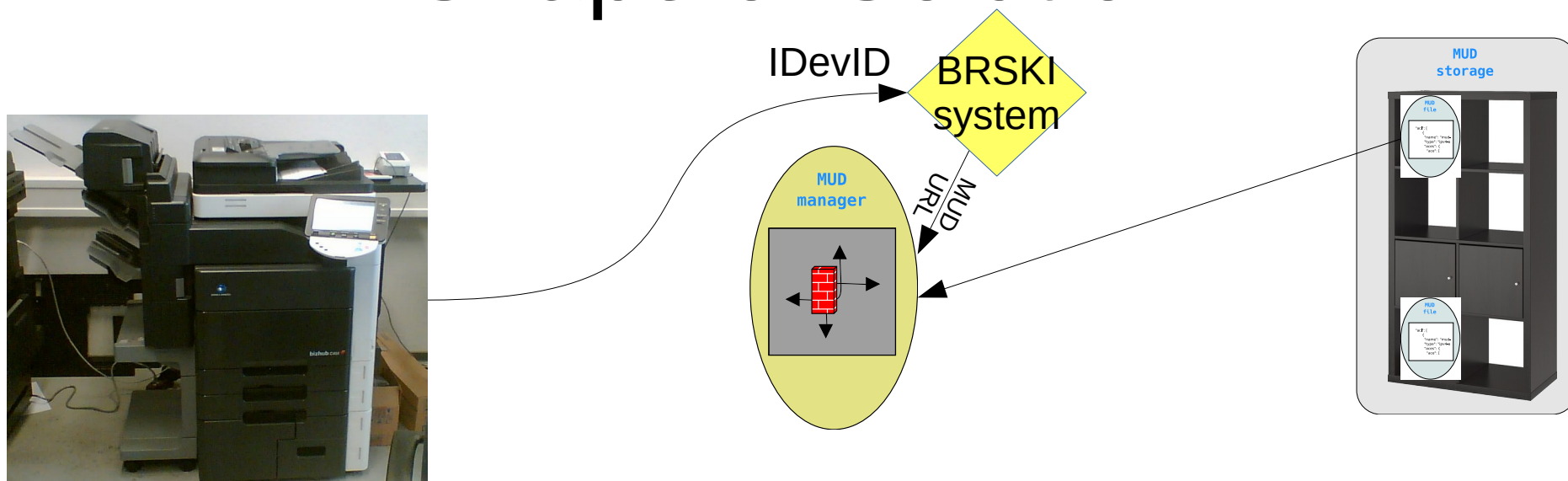


Shape of Solution



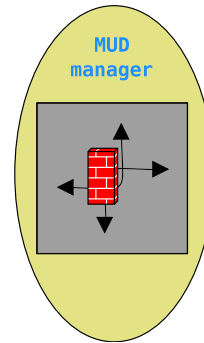
- IoT device tells MUD manager about it's URL
 - The first time, using IDevID
 - Or may TOFU on DHCP/LLDP

Shape of Solution



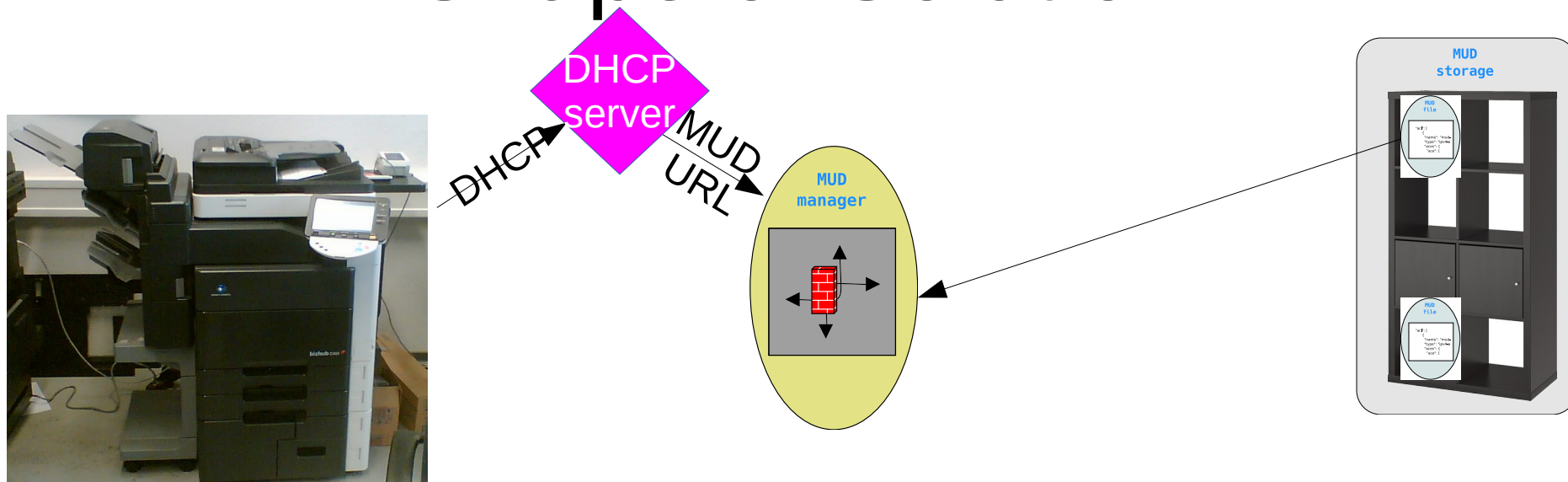
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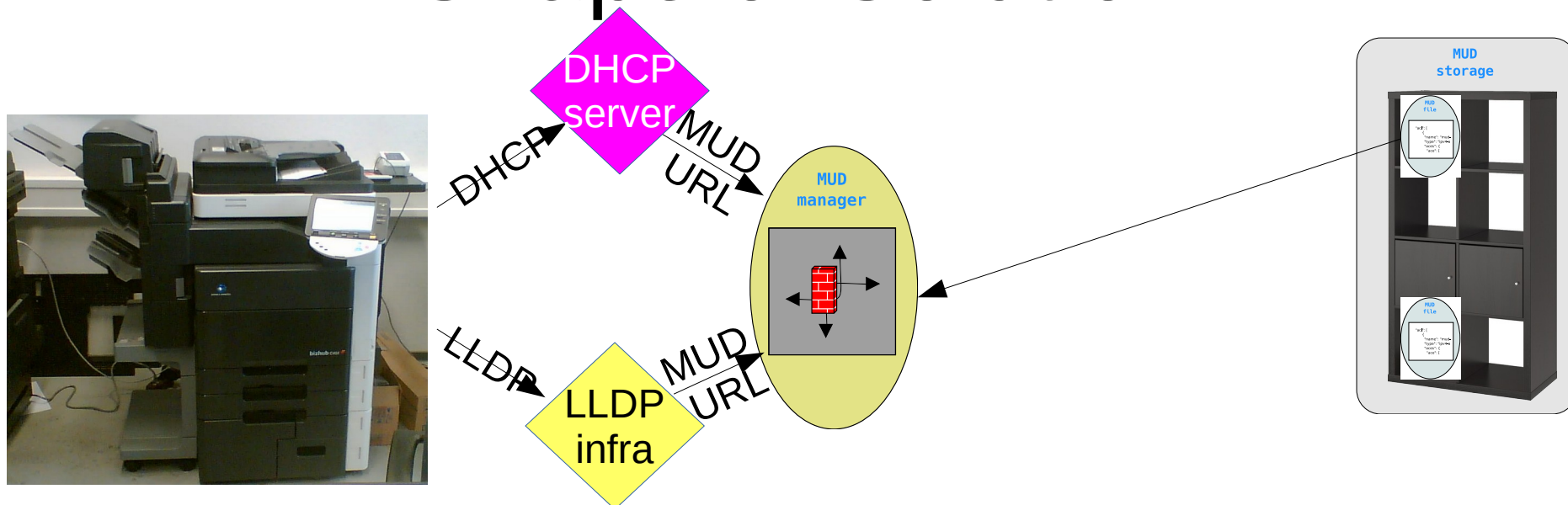
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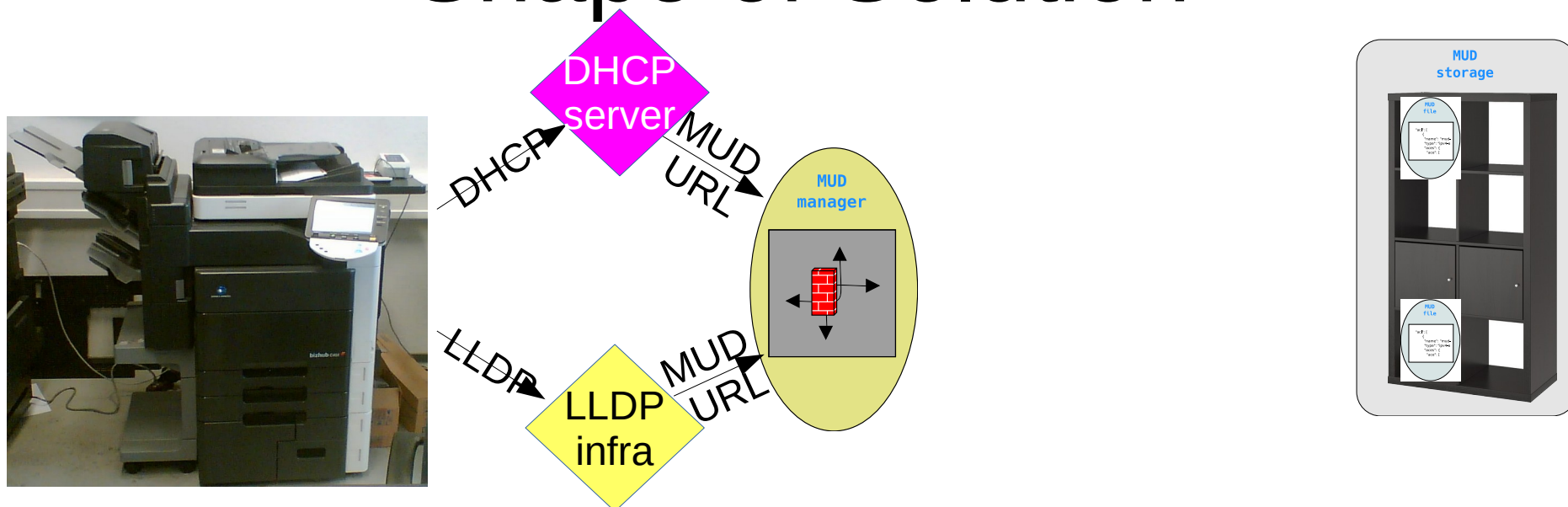
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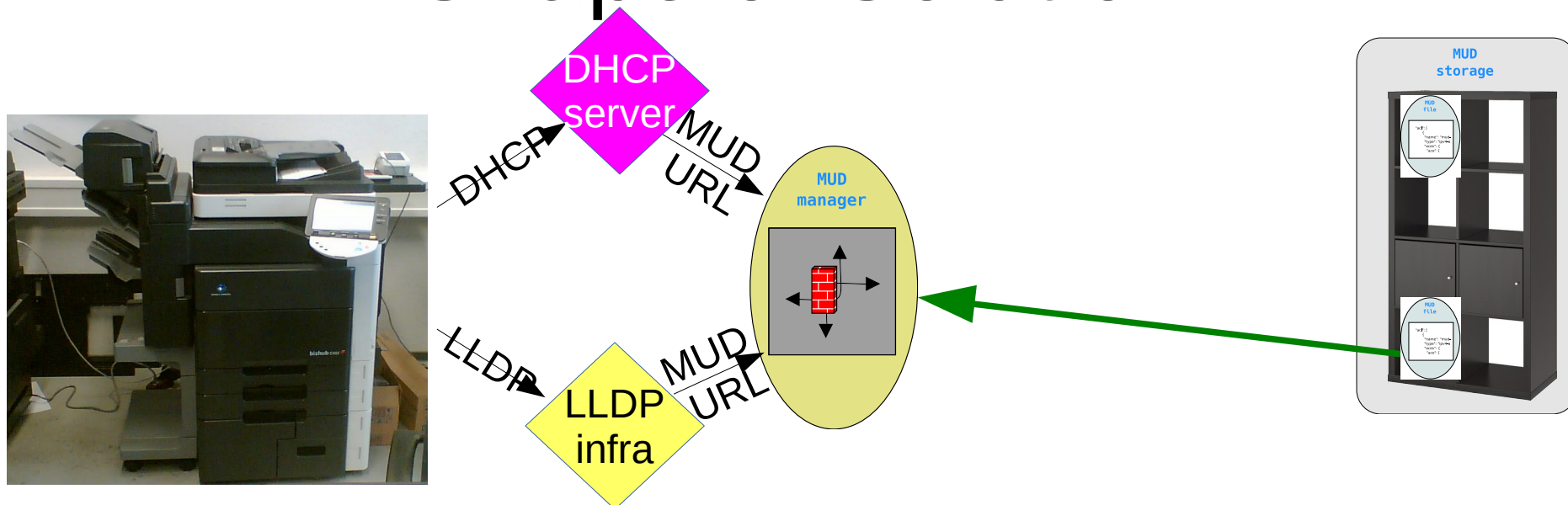
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Shape of Solution



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- Updates to URL would be restricted to the last component

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What kind of updates?

- Updates to URL would be restricted to the last component



Now what?



- WG adopted in January, 2021 into opsawg
- OPSAWG too busy, IOTOPS rechartered, document moved to IOTOPS in 2025
- Stable for awhile: WGLC
 - Any issue with WG document, standards track, updating ISE RFC8520 document?
- QUESTIONS?



Updates to IdevID?

- If IDevID is the most secure, why not update that?
 - Not easy to do for many products, IDevID is ideally stored in TPM
- If the IDevID can be updated, then can a malware update it too?